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Topic : Advance SQL Assignment

All tables (customer, order, item, category, color, store) were created and populated as per the given schema.

Sample records are shown below for verification.

253 • `SELECT * FROM customer;`

|   | id | first_name | last_name | favorite_color_id | city      |
|---|----|------------|-----------|-------------------|-----------|
| ▶ | 1  | Amit       | Sharma    | 1                 | Mumbai    |
|   | 2  | Priya      | Verma     | 2                 | Delhi     |
|   | 3  | Rahul      | Reddy     | 3                 | Hyderabad |
|   | 4  | Sneha      | Iyer      | 4                 | Chennai   |
|   | 5  | Vikas      | Singh     | 5                 | Lucknow   |
|   | 6  | Neha       | Gupta     | 6                 | Mumbai    |
|   | 7  | Arjun      | Mehta     | 7                 | Pune      |
|   | 8  | Kavya      | Nair      | 8                 | Kochi     |
|   | 9  | Rohit      | Patel     | 9                 | Ahmedabad |
|   | 10 | Pooja      | Das       | 10                | Kolkata   |

254 • `SELECT * FROM `order`;`

|   | id | customer_id | item_id | item_cnt | store_id | order_channel | day_dt     |
|---|----|-------------|---------|----------|----------|---------------|------------|
| ▶ | 1  | 1           | 1       | 2        | 1        | Store         | 2024-01-10 |
|   | 2  | 2           | 2       | 1        | 2        | Online        | 2024-01-11 |
|   | 3  | 3           | 3       | 3        | 3        | Store         | 2024-01-12 |
|   | 4  | 4           | 4       | 1        | 4        | Store         | 2024-01-13 |
|   | 5  | 5           | 5       | 2        | 5        | Online        | 2024-01-14 |
|   | 6  | 6           | 6       | 1        | 1        | Store         | 2024-01-15 |
|   | 7  | 7           | 7       | 2        | 6        | Store         | 2024-01-16 |
|   | 8  | 8           | 8       | 1        | 13       | Online        | 2024-01-17 |
|   | 9  | 9           | 9       | 3        | 8        | Store         | 2024-01-18 |
|   | 10 | 10          | 10      | 1        | 7        | Store         | 2024-01-19 |

255 • `SELECT * FROM item;`

|   | id | name           | size | price   | margin | color_id | category_id |
|---|----|----------------|------|---------|--------|----------|-------------|
| ▶ | 1  | Slim Fit Jeans | M    | 2000.00 | 300.00 | 1        | 4           |
|   | 2  | Regular Jeans  | L    | 1800.00 | 250.00 | 2        | 4           |
|   | 3  | Denim Jeans    | XL   | 2200.00 | 350.00 | 3        | 4           |
|   | 4  | Black Jeans    | M    | 2100.00 | 300.00 | 4        | 4           |
|   | 5  | White Jeans    | S    | 1900.00 | 200.00 | 5        | 4           |
|   | 6  | Blue T-Shirt   | M    | 800.00  | 100.00 | 2        | 5           |
|   | 7  | Green Shirt    | L    | 1200.00 | 150.00 | 3        | 6           |
|   | 8  | Formal Shirt   | XL   | 1500.00 | 200.00 | 4        | 6           |
|   | 9  | Casual Shirt   | M    | 1300.00 | 150.00 | 5        | 6           |
|   | 10 | Jacket         | L    | 2500.00 | 400.00 | 6        | 10          |

256 • `SELECT * FROM category;`

| Result Grid                                                                                                                                                                                                                                                 |    |              |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------------|-----------|
| Filter Rows: <input type="text"/>                                                                                                                                                                                                                           |    |              |           |
| Edit:    |    |              |           |
| Export/Import:                                                                        |    |              |           |
| Wrap Cell Content:                                                                                                                                                       |    |              |           |
|                                                                                                                                                                                                                                                             | id | name         | parent_id |
| ▶                                                                                                                                                                                                                                                           | 1  | mens         | NULL      |
|                                                                                                                                                                                                                                                             | 2  | womens       | NULL      |
|                                                                                                                                                                                                                                                             | 3  | kids         | NULL      |
|                                                                                                                                                                                                                                                             | 4  | mens_jeans   | 1         |
|                                                                                                                                                                                                                                                             | 5  | mens_tshirt  | 1         |
|                                                                                                                                                                                                                                                             | 6  | mens_shirts  | 1         |
|                                                                                                                                                                                                                                                             | 7  | womens_dress | 2         |
|                                                                                                                                                                                                                                                             | 8  | womens_tops  | 2         |
|                                                                                                                                                                                                                                                             | 9  | kids_wear    | 3         |
|                                                                                                                                                                                                                                                             | 10 | mens_jackets | 1         |

257 • `SELECT * FROM color;`

| Result Grid                                                                                                                                                                                                                                                 |    |        |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------|-----------|
| Filter Rows: <input type="text"/>                                                                                                                                                                                                                           |    |        |           |
| Edit:    |    |        |           |
| Export/Import:                                                                        |    |        |           |
| Wrap Cell Content:                                                                                                                                                       |    |        |           |
|                                                                                                                                                                                                                                                             | id | name   | extra_fee |
| ▶                                                                                                                                                                                                                                                           | 1  | Red    | 10.00     |
|                                                                                                                                                                                                                                                             | 2  | Blue   | 5.00      |
|                                                                                                                                                                                                                                                             | 3  | Green  | 0.00      |
|                                                                                                                                                                                                                                                             | 4  | Black  | 15.00     |
|                                                                                                                                                                                                                                                             | 5  | White  | 0.00      |
|                                                                                                                                                                                                                                                             | 6  | Yellow | 5.00      |
|                                                                                                                                                                                                                                                             | 7  | Pink   | 10.00     |
|                                                                                                                                                                                                                                                             | 8  | Orange | 5.00      |
|                                                                                                                                                                                                                                                             | 9  | Purple | 10.00     |
|                                                                                                                                                                                                                                                             | 10 | Grey   | 0.00      |

258 • `SELECT * FROM store;`

| Result Grid                                                                                                                                                                                                                                                       |    |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----------|
| Filter Rows: <input type="text"/>                                                                                                                                                                                                                                 |    |           |
| Edit:    |    |           |
| Export/Import:                                                                          |    |           |
| Wrap Cell Content:                                                                                                                                                           |    |           |
|                                                                                                                                                                                                                                                                   | id | city      |
| ▶                                                                                                                                                                                                                                                                 | 1  | Mumbai    |
|                                                                                                                                                                                                                                                                   | 2  | Delhi     |
|                                                                                                                                                                                                                                                                   | 3  | Hyderabad |
|                                                                                                                                                                                                                                                                   | 4  | Chennai   |
|                                                                                                                                                                                                                                                                   | 5  | Bangalore |
|                                                                                                                                                                                                                                                                   | 6  | Pune      |
|                                                                                                                                                                                                                                                                   | 7  | Kolkata   |
|                                                                                                                                                                                                                                                                   | 8  | Ahmedabad |
|                                                                                                                                                                                                                                                                   | 9  | Jaipur    |
|                                                                                                                                                                                                                                                                   | 10 | Lucknow   |

**Exercise 1 : Display the name of clothing items ( item name ), their color ( color name), and the last name and first name of the customer(s) who bought the apparel in their favorite color in 'Mumbai' city through store transactions in the 'mens\_jeans' category. Sort rows according to color, in ascending order.**

- First I joined all the required tables to get complete data.
- Then I applied conditions like city, category and channel in WHERE to filter step by step.
- checked favorite color condition to match customer preference.

```

195  -- Exercise 1
196  -- Clean and maintainable version
197  •  SELECT
198      i.name AS item_name,
199      co.name AS color_name,
200      cu.first_name,
201      cu.last_name
202  FROM customer cu
203  JOIN `order` o ON cu.id = o.customer_id
204  JOIN store s ON o.store_id = s.id
205  JOIN item i ON o.item_id = i.id
206  JOIN category ca ON i.category_id = ca.id
207  JOIN color co ON i.color_id = co.id
208  WHERE
209      s.city = 'Mumbai'
210      AND o.order_channel = 'Store'
211      AND ca.name = 'mens_jeans'
212      AND cu.favorite_color_id = i.color_id
213  ORDER BY co.name ASC;

```

- Same query written in a different way by putting some conditions inside JOIN.
- Just tried this approach to understand how filtering works during joins.

```

215  -- Same logic but filtering inside JOIN clauses
216  •  SELECT
217      i.name,
218      co.name,
219      cu.first_name,
220      cu.last_name
221  FROM customer cu
222  JOIN `order` o ON cu.id = o.customer_id
223  JOIN store s ON o.store_id = s.id AND s.city = 'Mumbai'
224  JOIN item i ON o.item_id = i.id AND cu.favorite_color_id = i.color_id
225  JOIN category ca ON i.category_id = ca.id AND ca.name = 'mens_jeans'
226  JOIN color co ON i.color_id = co.id
227  WHERE o.order_channel = 'Store'
228  ORDER BY co.name ASC;

```

- Output shows only those records which satisfy all conditions.

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

|   | name           | name | first_name | last_name |
|---|----------------|------|------------|-----------|
| ▶ | Slim Fit Jeans | Red  | Amit       | Sharma    |

## Exercise 2 : Select the last name and first name of customers and the name of their favorite color for customers with no purchases.

- Added one extra customer who has not placed any orders
- Used LEFT JOIN to include all customers
- If a customer has no orders, order id will be NULL
- So I filtered those rows to get customers with no purchases
- Output shows customers who have not made any purchases

```

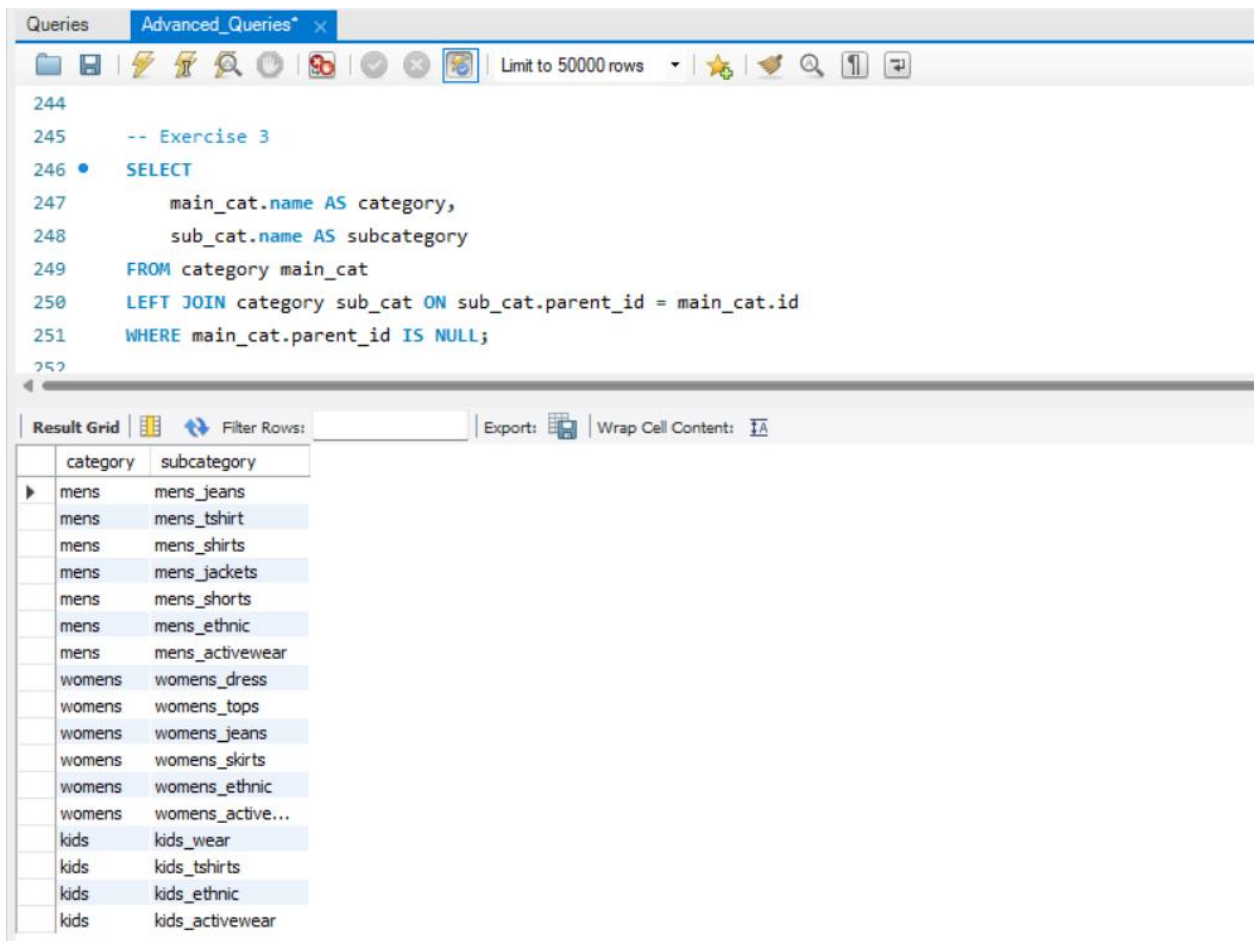
230 -- Inserting Test customer sample with no purchases
231 • INSERT INTO customer (id, first_name, last_name, favorite_color_id, city)
232 VALUES (999, 'Test', 'NoOrder', 3, 'Mumbai');
233
234 -- Exercise 2
235 -- Customers with no purchases
236 • SELECT
237     cu.first_name,
238     cu.last_name,
239     co.name AS favorite_color
240 FROM customer cu
241 LEFT JOIN `order` o ON cu.id = o.customer_id
242 JOIN color co ON cu.favorite_color_id = co.id
243 WHERE o.id IS NULL;
244
245 -- Exercise 3

```

| Result Grid | Filter Rows: | Export:   | Wrap Cell Content: |
|-------------|--------------|-----------|--------------------|
|             | first_name   | last_name | favorite_color     |
| ▶           | Test         | NoOrder   | Green              |

**Exercise 3 : Select the name of the main categories (which have a NULL in the parent\_id column) and the name of their direct subcategory (if one exists). Name the first column category and the second column subcategory.**

- Used LEFT JOIN on the same category table to connect parent and subcategory
- parent\_id links subcategory to its main category
- filtered only main categories using parent\_id IS NULL
- Output shows each main category with its subcategories



The screenshot shows a SQL query editor with a query window titled "Advanced\_Queries\*". The query is as follows:

```
244
245 -- Exercise 3
246 • SELECT
247     main_cat.name AS category,
248     sub_cat.name AS subcategory
249 FROM category main_cat
250 LEFT JOIN category sub_cat ON sub_cat.parent_id = main_cat.id
251 WHERE main_cat.parent_id IS NULL;
```

Below the query editor, the "Result Grid" tab is active, displaying the results of the query in a table with two columns: "category" and "subcategory".

|   | category | subcategory      |
|---|----------|------------------|
| ▶ | mens     | mens_jeans       |
|   | mens     | mens_tshirt      |
|   | mens     | mens_shirts      |
|   | mens     | mens_jackets     |
|   | mens     | mens_shorts      |
|   | mens     | mens_ethnic      |
|   | mens     | mens_activewear  |
|   | womens   | womens_dress     |
|   | womens   | womens_tops      |
|   | womens   | womens_jeans     |
|   | womens   | womens_skirts    |
|   | womens   | womens_ethnic    |
|   | womens   | womens_active... |
|   | kids     | kids_wear        |
|   | kids     | kids_tshirts     |
|   | kids     | kids_ethnic      |
|   | kids     | kids_activewear  |